

# Junjie Nian 年骏杰


B.Eng. in Artificial Intelligence, Fudan University · Exchange Student at UC San Diego (Fall 2026)

EMAIL junjienian8@gmail.com PHONE (+86) 19370555931

GITHUB github.com/JunjieNian WEBSITE junjienian.com

PROFILES Google Scholar · DBLP · Semantic Scholar · ACL Anthology · ORCID


## ABOUT

Undergraduate researcher in Artificial Intelligence at Fudan University and exchange student at UC San Diego (Fall 2026). At FudanNLP ALEX Lab, advised by Prof. Yixin Cao, I study reasoning processes, agent reliability, and mechanistic interpretability.

## EDUCATION

**2024.9 – Present** — **Fudan University** — B.Eng. in Artificial Intelligence
School of Computer Science.

**Fall 2026** — **UC San Diego** — Exchange Student
Computer Science & Engineering.

**2021.9 – 2024.6** — **The High School Affiliated to Fudan University**

## RESEARCH EXPERIENCE

**2025.10 – Present** — **FudanNLP ALEX Lab** — Undergraduate Researcher
Advised by Prof. Yixin Cao. Research on Large Reasoning Model capabilities, agent systems, and mechanistic interpretability.

**2025.3 – 2025.6** — **Embodied Intelligence Research Project** — Tengfei Innovation Program
Advised by Prof. Shang Li. Built small-scale robots with asynchronous servo motor control and embedded development board integration.

## PUBLICATIONS

Kang Chen*, **Junjie Nian***, Yixin Cao, Yugang Jiang. "Disagree to Explore, Agree to Commit: Routing-Guided Test-Time Scaling for Software Agents." **Preprint, 2026. [arXiv] [Code]**

Kang Chen, Minshen Yu, **Junjie Nian**, Yaoning Wang, Yixin Cao, Yugang Jiang. "Does the Same Token Mean the Same State? MoE Routing as Signal for Reasoning Control." **Preprint, 2026. [arXiv] [Website]**

**Junjie Nian***, Kang Chen*, Ge Zhang, Yixin Cao, Yugang Jiang. "TraceGraph: Shared Decision Landscapes for Diagnosing and Improving Agent Trajectories." **Preprint, 2026. [arXiv] [Code] [Trajectory Lens]**

Kang Chen*, **Junjie Nian***, Yixin Cao, Yugang Jiang. "SliceGraph: Mapping Process Isomers in Multi-Run Chain-of-Thought Reasoning." **Preprint, 2026. [arXiv] [Code]**

Kang Chen*, Zhuoka Feng*, Sihan Zhao, Kai Xiong, **Junjie Nian**, Yaoning Wang, Changyi Xiao, Yixin Cao. "NEX: Neuron Explore-Exploit Scoring for Label-Free Chain-of-Thought Selection and Model Ranking." **Preprint, 2026. [arXiv]**

Kang Chen*, Fan Yu*, **Junjie Nian**, Sihan Zhao, Zhuoka Feng, Zijun Yao, Heng Wang, Minshen Yu, Yixin Cao. "Thinking Traps in Long Chain-of-Thought: A Measurable Study and Trap-Aware Adaptive Restart." **ACL 2026 Findings. [arXiv] [ACL Anthology]**

Zhuoka Feng*, Kang Chen*, Sihan Zhao†, Kai Xiong†, Yaoning Wang†, Minshen Yu, **Junjie Nian**, Changyi Xiao, Yixin Cao, Yugang Jiang. "ARM: Role-Conditioned Neuron Transplantation for Training-Free Generalist LLM Agent Merging." **EMNLP 2026 Oral. [arXiv] [Website]**

* Equal contribution † Equal second-author contribution

## HONORS & AWARDS

| | |
|---|---|
| Fudan University Outstanding Student Scholarship | **2026** |
| Duckbill Special Scholarship, Fudan University | **2025** |
| Third Prize, China Undergraduate Mathematical Contest in Modelling, Shanghai | **2025** |
| Third Prize, The Chinese Mathematics Competitions for College Students, Shanghai | **2025** |
| Freshman Scholarship, Fudan University | **2025** |
| First Prize, Tengfei College Entrance Exam Freshman Scholarship, Fudan University | **2024** |

## SKILLS

**Programming:** Python, C/C++, JavaScript
**Machine Learning & Scientific Computing:** PyTorch, NumPy, SciPy
**Web Development:** React, FastAPI, Three.js, HTML/CSS
**Tools:** Git, Docker, LaTeX

## ENGLISH PROFICIENCY

TOEFL **111** CET-6 **619** CET-4 **636**